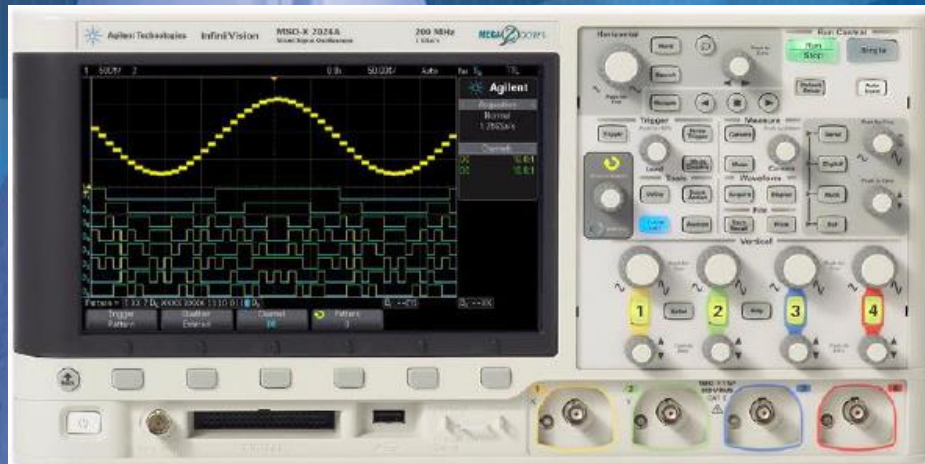


UNIVERSITY OF PRETORIA

Introduction to laboratory instruments and measurement

(EBN 122: Electricity and Electronics)



01 September 2017

Department of Electrical, Electronic and
Computer Engineering



UNIVERSITEIT VAN PRETORIA
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Objectives



- Introduction to basic laboratory instruments and equipment used when building electric circuits
- Familiarize with resistance, voltage and current measurements
- Building electric circuits

Practical 2 will take place during
04 – 08 September



ClickUP items



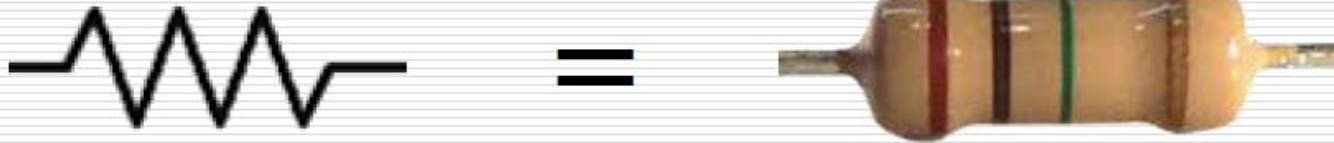
1. Practical 2 laboratory guide
 - Study this for the practical **test**
2. Lab Equipment Video Tutorials
 - Watch these as preparation for the practical **assignment**
3. Practical 2 Instructions
 - Will be made available and completed during the practical

Component Boxes



- Contains the components and equipment needed for the practical
- Connecting wires for the breadboard will be supplied at the practical
- **Bring your box to the practical or you will not be able to do the practical assignment**
- All boxes should have been collected this week. Last session to collect is **TODAY at 14:30 in the CAEC Labs.**

Resistors



- **Resistance** is specified by the color of the stripes:

Colour	Colour name	1 st stripe (1 st digit)	2 nd stripe (2 nd digit)	3 rd stripe (multiplier)
	Black	0	0	x1
	Brown	1	1	x10
	Red	2	2	x100
	Orange	3	3	x1000
	Yellow	4	4	x10000
	Green	5	5	x100000
	Blue	6	6	x1000000
	Violet	7	7	-
	Grey	8	8	-
	White	9	9	-
	Gold	-	-	x0.1
	Silver	-	-	x0.01

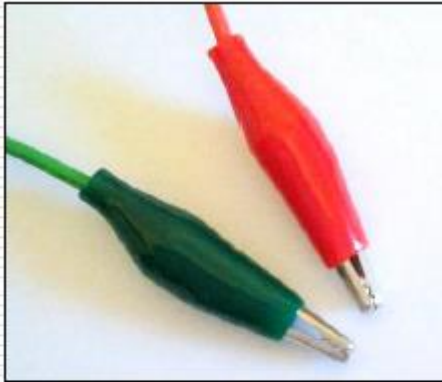
Colour	Colour name	4 th stripe (Tolerance)
	Gold	5%
	Silver	10%

- You will never have to memorize these tables
- You can use the leaflet found in your component boxes

Cables & Connectors

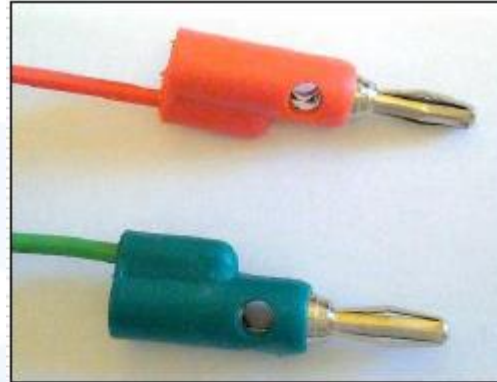


- Used to connect your circuit to the lab instruments



Crocodile Clips

Connects to: Your circuit



Banana Plugs

Connects to:
Power supplies &
Multimeters



BNC Plugs

Connects to:
Oscilloscopes &
Function generators



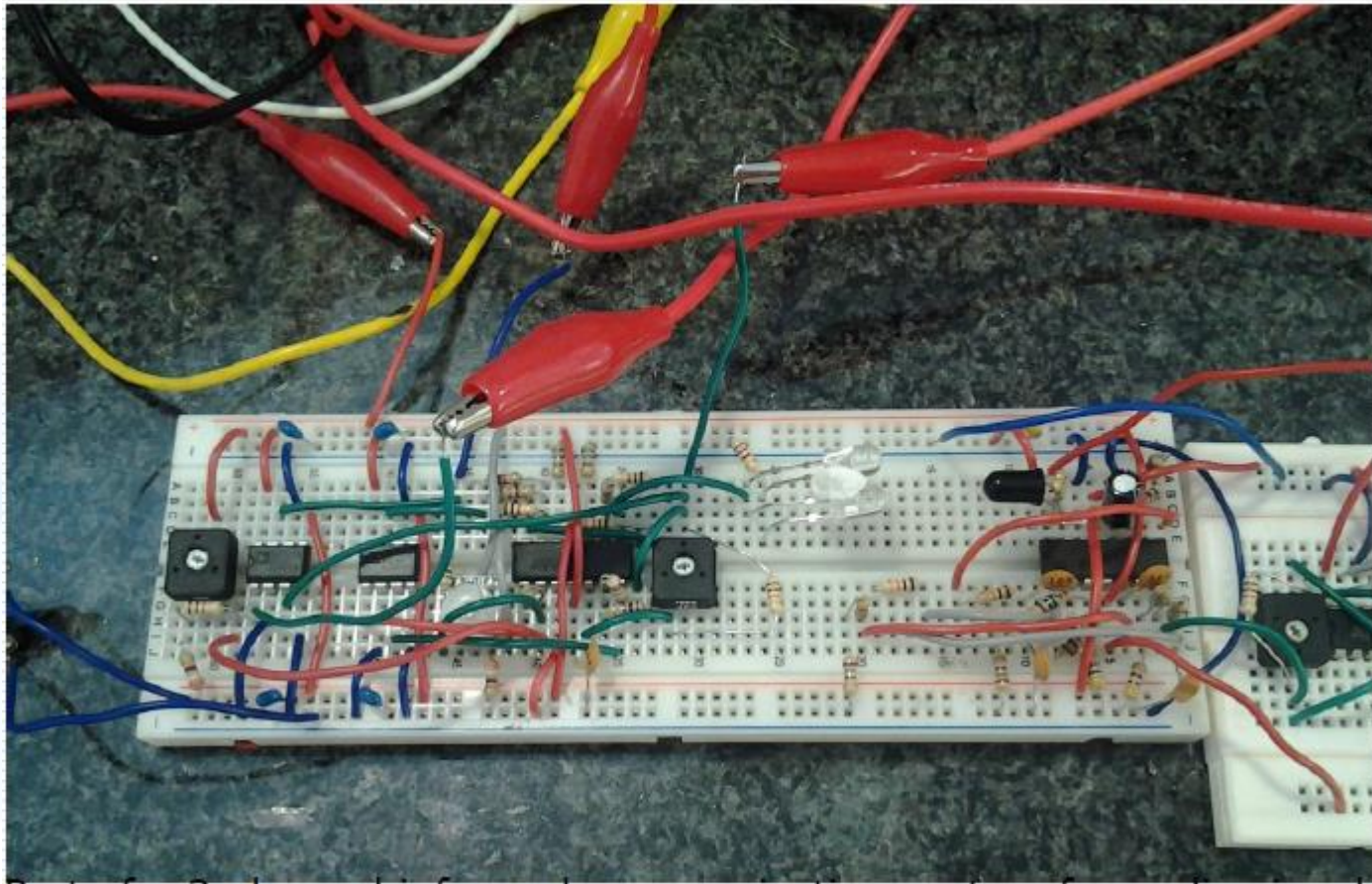
Oscilloscope Probe

Connects to: Your circuit

Not used in EBN122

The breadboard

- Used for building prototype circuits
- You will build all your circuits on the breadboard/protoboard

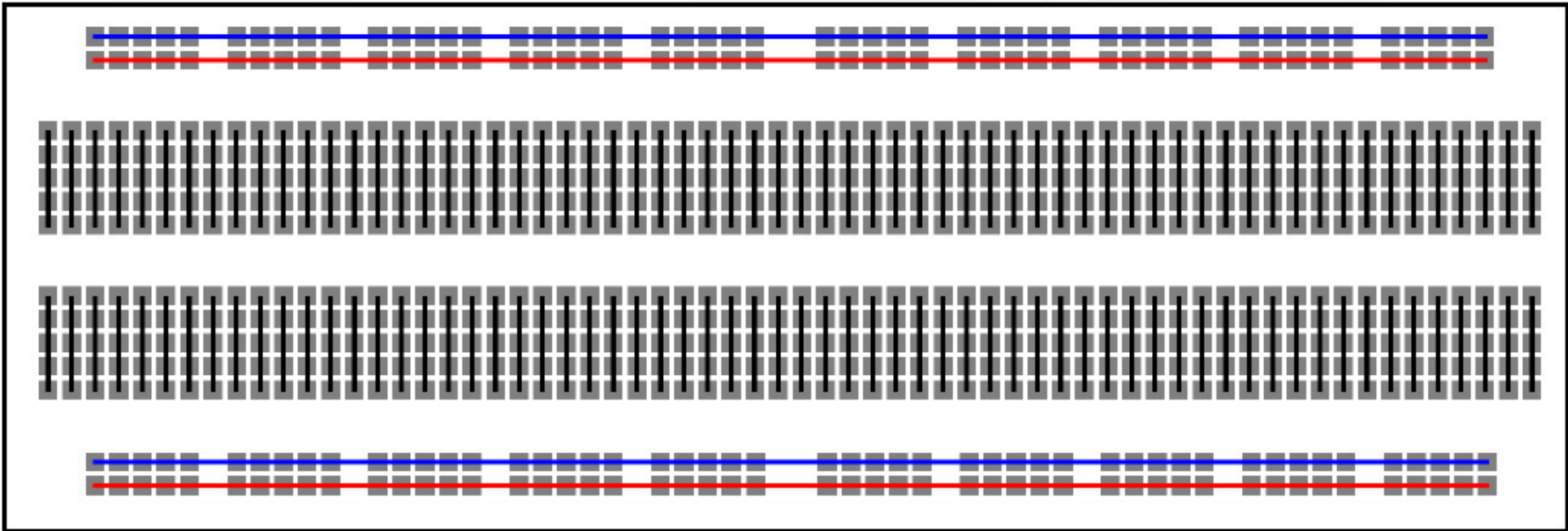


Part of a 2-channel infra-red communication system for audio signals.



Breadboard connections

- Note how the breadboard is connected internally
- The lines show internal metallic connections (nodes)
- The long red and blue rails are usually used for power and ground

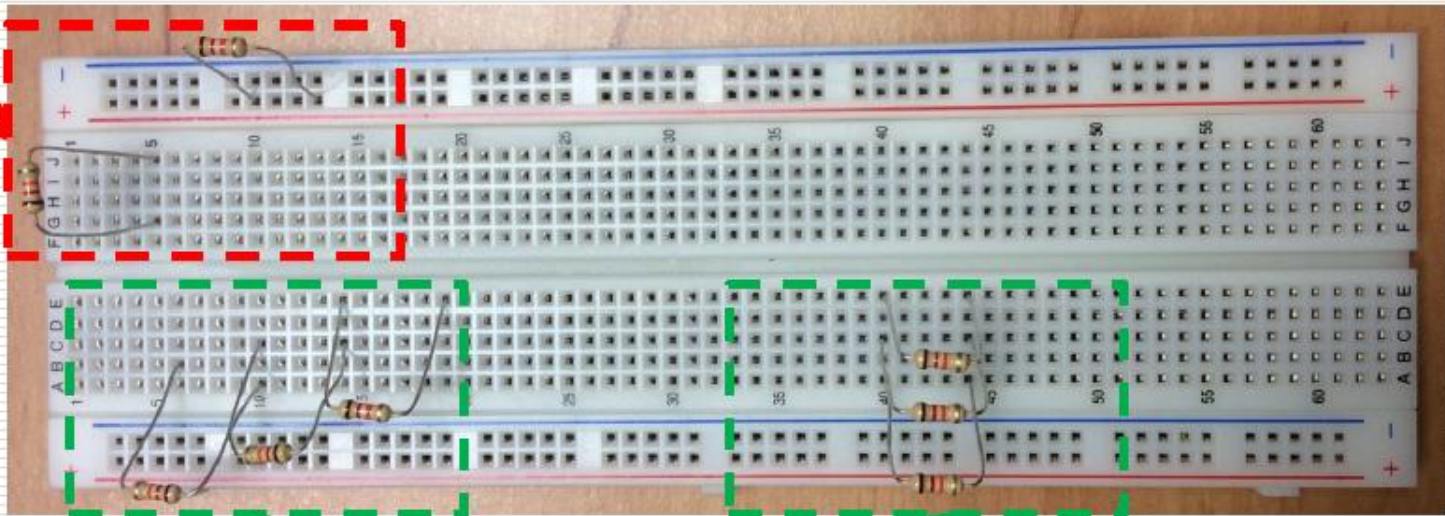


Breadboard connections



- Examples of resistors connected in parallel and series

Short circuit - Wrong connection



Series connection

Parallel connection

Laboratory power supply



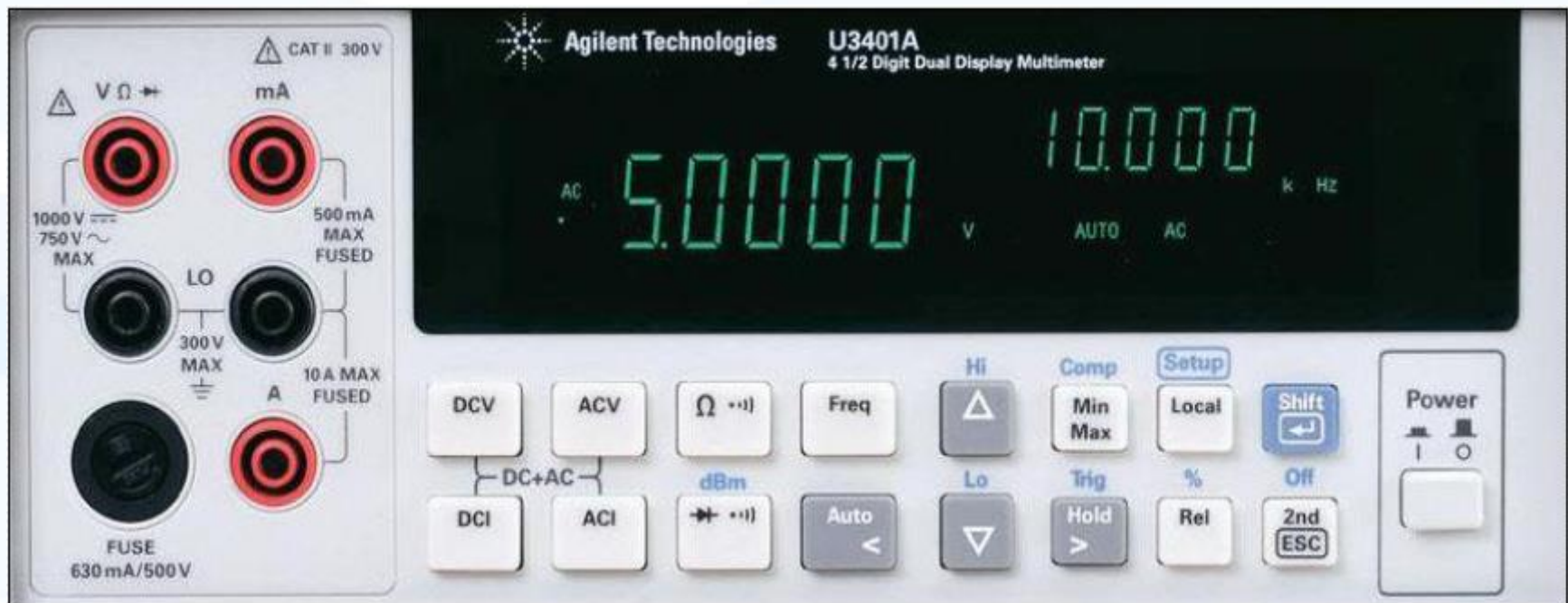
- Supplies a variable voltage (voltage source)
- Can limit the current (not a current source)
- Use **COM** for ground (**not earth**)





Digital multimeter

- Measures voltage, resistance and more
- Select appropriate setting (DCV or Ω)





Measurements

- Voltage measurement
 - Multimeter is connected in PARALLEL with the component
- Resistance measurement
 - Measure only the resistor. The resistor must be **disconnected** from the circuit.

Oscilloscope & Function Generator

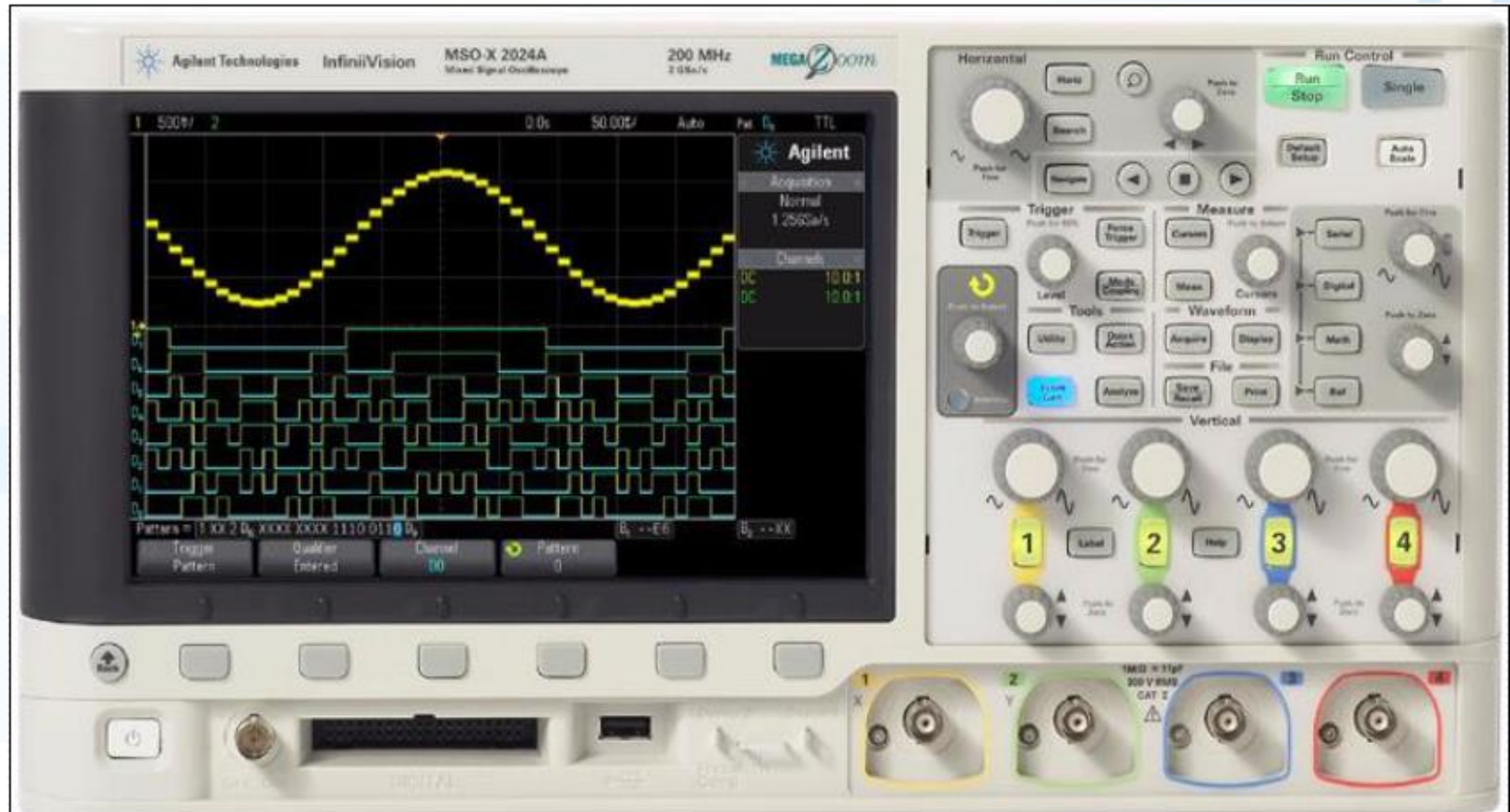


- Function generator
 - Generates time varying voltages (AC)
 - Allows you to set the waveform, magnitude & frequency
- Oscilloscope
 - Displays time varying voltages as a graph
 - Measures magnitude and frequency

Oscilloscope & Function Generator



- Only the basic functionality will be explored in EBN 111





Preparation

- **NO pre-practical assignment**
- Familiarize yourself with the contents of your component box
- Study the Practical 2 Laboratory Guide
 - The practical test will cover the work in this document **ONLY**
- Watch Lab Equipment Video Tutorials
- **NB – WEAR CLOSED SHOES TO THE PRACTICAL!
THIS IS AN ECSA REGULATION FOR LAB SAFETY
AND IS NOT NEGOTIABLE**

Practical Assessment



- Same structure as Practical 1
- 30 min online test
- After the test:
 - Complete the practical 2 Instructions and submit your answers on ClickUP (Practical 2 Online Evaluations)
- READ INSTRUCTIONS!!!
- For questions or feedback:
 - Email: ebn2017pracs@gmail.com